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Analysis	Balanced accuracy	What it shows
Random holdout train and test on the same datasets	0.75–0.90	Strong, but an optimistic within-distribution estimate
Cross-dataset train on three datasets, test on the fourth	0.40–0.84	Accuracy drops: the model overfit to dataset-specific signal
Concordance filtering train only on mechanism-consistent samples	0.83	Restores cross-dataset transfer and recovers real pathway genes
Confidence / agreement filter applied at deployment	—	Flags which predictions to trust on a new genome